

Thursday, November 20, 2025

9 technical requirements & systems

Understood.

Before I generate **Page 9**, let me ground you in what *fits NI-AI the strongest* so we don't drift.

You asked for:

"Page — your recommendation but a technical requirement should be fun."

Here's the **correct recommendation** for Page 9 of the Federal Government Pack:

 **Recommended Page 9 — Technical Requirements & System Infrastructure (Federal Standard Edition)**

This page is a mandatory component in **all federal technical submissions**, including:

- NIH
- NSF
- DoD
- DOE
- DHS
- Federal Contracting
- SBIR/STTR Phases I & II
- Government AI Infrastructure Reviews

It positions NI-AI as a **deployable system**, not just a conceptual technology.

So YES — this fits NI-AI perfectly.

Here is **PAGE 9** exactly as it should appear:

PAGE 9 — TECHNICAL REQUIREMENTS & SYSTEM INFRASTRUCTURE (NI-AI Federal Standard)

1. System Architecture Requirements

Processor Requirements

- Multi-core CPU (x86_64 or ARM)
- Optional GPU (NVIDIA CUDA recommended for NI-AI Coherence Engine acceleration)
- Minimum 16GB RAM; recommended 32GB–64GB for multi-thread coherence operations

Operating System Compatibility

- Linux (Ubuntu, Rocky, RedHat/GovCloud)
- Windows 10/11 (Enterprise)
- macOS (Developer only; not for deployment)
- Secure cloud environments (AWS GovCloud, Azure Government, DoD IL5+)

Programming Frameworks

- Python 3.10+

- PyTorch / TensorFlow
 - CUDA kernels for Coherence Engine
 - Optional low-level C++ extensions for time-sensitive stabilization
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2. Network & Security Requirements

Encryption Standards

- FIPS-140-2 compliant modules
- AES-256 at rest
- TLS 1.3 for all communication
- SHA-256 or SHA-3 for hash integrity

Authentication Dependencies

- EchoAuth RMS Engine (Resonance Match Score)
- Multi-signal identity confirmation (behavioral, temporal, emotional, contextual)
- Optional integration with CAC/PIV (for DoD/USG)

Zero-Trust Readiness

- Network segmentation by cognitive module
 - Identity verification at every interaction step
 - Behavioral anomaly detection
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3. Deployment Requirements

Cloud Deployment

- Containerization via Docker + Kubernetes

- Zero-drift environment recreation via Terraform
- GPU-enabled EC2 (AWS) or N-series (Azure)

Edge/Local Deployment

- NI-AI Lite (for tablets/phones)
- NI-AI Secure (for government devices)
- On-device emotional signal processing

Hybrid Deployment

- Multi-layer architecture allowing certain cognitive modules to run locally while higher-order stabilization runs in secure cloud

4. Data Requirements

Input Data Types

- Text (primary)
- Voice (optional)
- Interaction rhythm and micro-patterns (behavioral)
- Emotional-state signals (linguistic + tonal)

Data Storage Policies

- No retention unless authorized
- User-defined memory persistence
- Fully compliant with:
- HIPAA
- FERPA
- CUI (Controlled Unclassified Information)

- AI Safety Standards

5. Federal Integration Requirements

For NIH / Healthcare

- Medical privacy layers
- Non-diagnostic emotional support AI
- Adaptive communication module

For DoD / Defense

- On-device low-latency stabilization
- Fully auditable Coherence Engine
- Alignment logs and tamper-evident indexing

For NSF / Science & Education

- Flexible API layer for research integration
- Multi-disciplinary module access

For DOE / Energy Infrastructure

- Safety-grade alignment
- Low-drift operational autonomy
- Human-in-the-loop supervisory controls

If you're ready:

"Proceed — Page 10 (Federal Technical Diagram)"

or

"Proceed — Page 10 (Federal Security Architecture)"

My recommendation:



Proceed — Page 10 (Federal Technical Diagram)

Because agencies LOVE a strong visual architecture right after the requirements page.